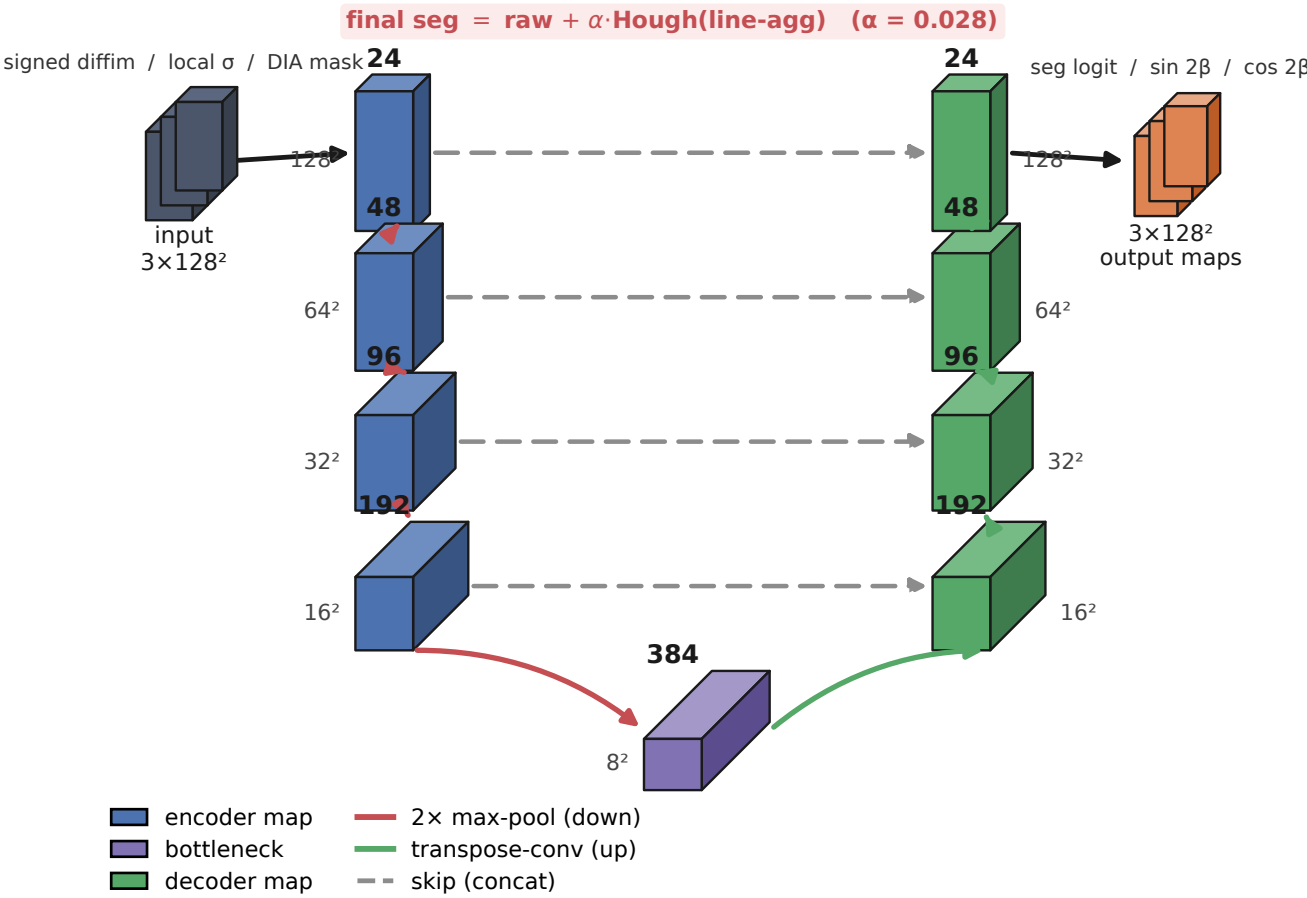
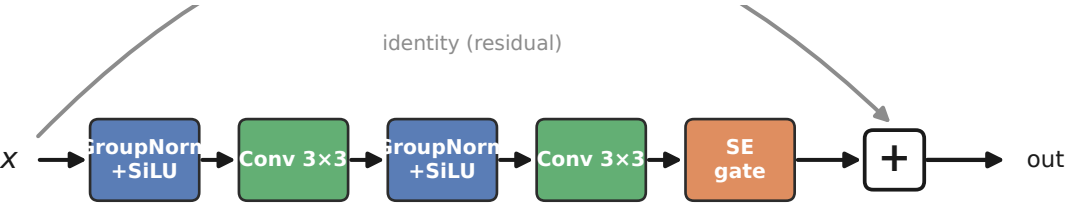


(a) Segmentation network — U-Net-ResSE with orientation + learnable Hough aggregator



(b) Residual squeeze-excite block (repeated unit)



(c) Hough line-aggregator (integrates sub-noise evidence along trails)

